

$$v = 334 \text{ m/s}$$

30.

Ans: C

For sample X: $A = \lambda_A N$

For sample Y: $3A = \lambda_Y N$

and since for half life $\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

we can combine the equations as follows:

$$A = \frac{\ln 2}{t_{1/2(X)}} N$$

$$3A = \frac{\ln 2}{t_{1/2(Y)}} N$$

$$\frac{3A}{A} = \frac{\frac{\ln 2}{t_{1/2(Y)}} N}{\frac{\ln 2}{t_{1/2(X)}} N}$$

$$3 = \frac{t_{1/2(X)}}{t_{1/2(Y)}}$$

